

GRIDGUARD AI 4.0

National Smart Grid Executive Intelligence Report

Generated: 2026-02-07T23:29:23.694Z
Scope: National operational risk, theft intelligence, transformer reliability, alert posture, and recommended actions.
Source: Live platform data (regions, alerts, transformer telemetry, meter anomalies, and reading streams).

Executive Narrative

This executive brief presents a formal live operational assessment of GRIDGUARD AI 4.0 based on current telemetry, anomaly outputs, and alert intelligence.

At the time of generation, national average theft score stands at 20.51% and national distribution loss stands at 6.9%. The transformer risk index is 42.18% and the carbon optimization score is 70.43%.

System footprint currently includes 35 regions, 120 transformers, and 1080 smart meters.

Risk zoning currently indicates 0 red zones, 0 yellow zones, and 35 green zones.

The estimated daily financial impact of loss is INR 31,176.41 based on recent load behavior and live loss conditions.

Executive Summary:

- National loss remains elevated and requires feeder-level loss reduction focus.
- Transformer risk is moderate with pockets of overload stress.
- Theft exposure is concentrated in a subset of high-density regions.
- Prioritize vigilance dispatch, load balancing, and preventive transformer maintenance.

Key Metrics

National Theft Score (Avg): 20.51%
National Distribution Loss (Avg): 6.9%
Transformer Risk (Avg): 42.18%
Carbon Optimization (Avg): 70.43%
Open Alerts: 70
High Alerts (24h): 25
Estimated Daily Financial Impact (INR): 31,176.41

National Operational Snapshot

Total Regions Monitored: 35
Total Transformers Monitored: 120
Total Smart Meters Monitored: 1080
National Theft Score (Avg): 20.51%
National Distribution Loss (Avg): 6.9%
National Transformer Risk (Avg): 42.18%
National Carbon Optimization Score (Avg): 70.43%

Risk Zone Distribution

Red Zones (>70 theft score): 0
Yellow Zones (40-70 theft score): 0
Green Zones (<40 theft score): 35

Alert Intelligence

Total Alerts Logged: 77

Open Alerts (Unacknowledged): 70

High Severity Alerts in Last 24h: 25

- [2026-02-07T23:29:11.979Z] (MEDIUM) Maharashtra: Anomaly detected for meter f12ecf98-21e9-45ed-ba9b-797d93d5a695
- [2026-02-07T23:29:02.925Z] (HIGH) Gujarat: Anomaly detected for meter b330a3de-aea1-4bfb-94c9-7ca178c5fe7d
- [2026-02-07T23:28:35.773Z] (MEDIUM) Delhi: Anomaly detected for meter 703fb42d-e2bf-4299-a10c-60cf4dfe2924
- [2026-02-07T23:28:20.675Z] (HIGH) Tamil Nadu: Anomaly detected for meter 89b31c63-5cec-43df-b7ee-b6246607da98
- [2026-02-07T23:27:59.540Z] (MEDIUM) Assam: Anomaly detected for meter 79bd2551-024c-4a1d-8ea4-a8842ab4d7fe
- [2026-02-07T23:27:38.401Z] (MEDIUM) Tamil Nadu: Anomaly detected for meter 6da49627-e458-40b4-a09a-028ec240f4ea
- [2026-02-07T23:26:53.172Z] (MEDIUM) Andaman and Nicobar: Anomaly detected for meter a305b225-2ec3-448c-8c45-7f016970340d
- [2026-02-07T23:26:38.085Z] (HIGH) Gujarat: Anomaly detected for meter 81ba998c-1b01-43f1-af09-6e9a593b825c
- [2026-02-07T23:26:32.052Z] (MEDIUM) Delhi: Anomaly detected for meter 703fb42d-e2bf-4299-a10c-60cf4dfe2924
- [2026-02-07T23:25:31.664Z] (MEDIUM) Chandigarh: Anomaly detected for meter d3608df5-d0c1-4a9e-9c51-9e988bf53504
- [2026-02-07T23:24:52.412Z] (MEDIUM) Maharashtra: Anomaly detected for meter c40ce289-7d12-493d-89e2-aaa0d5205ee7
- [2026-02-07T23:24:43.354Z] (HIGH) Andaman and Nicobar: Anomaly detected for meter 4f914240-e3e7-4c21-8370-1e3baa9270c1

Top High-Risk Regions

- 1. Mizoram | Theft 34% | Loss 11.29% | Transformer Risk 43.27% | Carbon 59% | Updated 2026-02-07T22:17:27.901Z
- 2. West Bengal | Theft 34% | Loss 6.71% | Transformer Risk 41.29% | Carbon 73% | Updated 2026-02-07T22:17:27.901Z
- 3. Sikkim | Theft 33% | Loss 5.17% | Transformer Risk 48.68% | Carbon 85% | Updated 2026-02-07T22:17:27.901Z
- 4. Himachal Pradesh | Theft 32% | Loss 9.67% | Transformer Risk 27.63% | Carbon 64% | Updated 2026-02-07T22:17:27.901Z
- 5. Dadra and Nagar Haveli | Theft 30% | Loss 4.91% | Transformer Risk 23.51% | Carbon 62% | Updated 2026-02-07T22:17:27.901Z
- 6. Jharkhand | Theft 29% | Loss 6.51% | Transformer Risk 45.79% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z
- 7. Madhya Pradesh | Theft 26% | Loss 10.36% | Transformer Risk 34.43% | Carbon 70% | Updated 2026-02-07T22:17:27.901Z

- 8. Orissa | Theft 25% | Loss 7.43% | Transformer Risk 63.02% | Carbon 67% | Updated 2026-02-07T22:17:27.901Z
- 9. Delhi | Theft 25% | Loss 8.25% | Transformer Risk 33.2% | Carbon 70% | Updated 2026-02-07T23:28:53.868Z
- 10. Gujarat | Theft 24% | Loss 8.03% | Transformer Risk 70.01% | Carbon 70% | Updated 2026-02-07T23:29:21.031Z

Top Vulnerable Transformers

- 1. Maharashtra | Transformer f695c28f | Load 74.59% | Temp 61.96 C | Health 0.38 | Risk YELLOW
- 2. Tamil Nadu | Transformer f833d1ba | Load 74.28% | Temp 52.74 C | Health 0.38 | Risk GREEN
- 3. Bihar | Transformer bb9efdb5 | Load 73.2% | Temp 51.02 C | Health 0.39 | Risk GREEN
- 4. Tamil Nadu | Transformer 8f9fdda4 | Load 70.89% | Temp 46.54 C | Health 0.41 | Risk GREEN
- 5. Gujarat | Transformer c8f45ace | Load 70.03% | Temp 48.68 C | Health 0.42 | Risk GREEN
- 6. Gujarat | Transformer 69d2c8b3 | Load 70.01% | Temp 51.62 C | Health 0.42 | Risk GREEN
- 7. Delhi | Transformer 67102a47 | Load 68.68% | Temp 62.66 C | Health 0.43 | Risk GREEN
- 8. Chandigarh | Transformer b776a51c | Load 68.67% | Temp 69.49 C | Health 0.43 | Risk YELLOW
- 9. Delhi | Transformer 0ed6262c | Load 68.45% | Temp 46.13 C | Health 0.43 | Risk GREEN
- 10. Andhra Pradesh | Transformer 9ab80d8c | Load 65.48% | Temp 61.41 C | Health 0.45 | Risk GREEN

Top Meter Anomalies (7-day window)

- 1. Arunachal Pradesh | Meter 13821bee-947 | Theft Probability 84% | Class RED | Detected 2026-02-07T23:20:42.183Z
- 2. Gujarat | Meter 81ba998c-1b0 | Theft Probability 83% | Class RED | Detected 2026-02-07T23:26:38.082Z
- 3. Delhi | Meter 9657afa8-a5b | Theft Probability 83% | Class RED | Detected 2026-02-07T21:50:38.095Z
- 4. Gujarat | Meter 375a665d-c07 | Theft Probability 82% | Class RED | Detected 2026-02-07T22:36:28.438Z
- 5. Karnataka | Meter 417f1359-105 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:42:26.999Z
- 6. Tamil Nadu | Meter 98223e50-2a8 | Theft Probability 81% | Class RED | Detected 2026-02-07T22:36:04.297Z
- 7. Maharashtra | Meter d8ef16cf-5fa | Theft Probability 80% | Class RED | Detected 2026-02-07T22:55:52.712Z
- 8. Andaman and Nicobar | Meter a69b1691-f8d | Theft Probability 79% | Class RED | Detected 2026-02-07T23:21:54.604Z
- 9. Delhi | Meter c3329c11-04b | Theft Probability 79% | Class RED | Detected 2026-02-07T22:08:09.428Z
- 10. Tamil Nadu | Meter 44e5eb02-9fd | Theft Probability 79% | Class RED | Detected 2026-02-07T22:00:21.340Z

24-Hour Load Trend

Average Power (24h): 2.49 kW

Estimated Daily Energy Loss: 4453.77 kWh

- 2026-02-07T21:30:00.000Z | Avg Power 2.807 kW | Readings 781
- 2026-02-07T22:30:00.000Z | Avg Power 2.173 kW | Readings 354

Recommended Actions

- Maintain active anomaly surveillance and sustain monthly anti-theft audit cadence in medium-risk states.
- Current national loss is within an acceptable band; continue preventive maintenance and loss monitoring.
- Initiate a 14-day transformer reliability sprint focused on high-load assets with low health index.
- Establish an alert triage war-room with SLA-based acknowledgement and closure tracking.
- Use LLM assistant for weekly executive briefings, including state-wise risk rationale and recommended field actions.

Methodology Note

This report is generated from live PostgreSQL operational data and near-real-time simulator/ingest streams.

Risk zone classification follows GRIDGUARD thresholds based on theft probability and transformer stress signals.

Financial impact estimation uses current average power, meter coverage, and loss percentage with an indicative INR conversion factor.

Generated by GRIDGUARD AI 4.0 automated reporting engine.